



**Tshwane University  
of Technology**

*We empower people*

**INSTRUCTIONS TO  
CANDIDATES**

1. All examination rules stated by the Tshwane University of Technology apply.
2. Ensure a single final version of your source code is handed in as requested.
3. State all necessary assumptions clearly in code commentary where needed.

**MARKS:** 50

**PAGES:** 15 (incl. cover)

**EXAMINER:**

Mr A.J. Smith

Mr M. Chauke

Prof J.A. Jordaan

**MODERATOR:**

Mr D Engelbrecht

**TIME:**

90 Minutes

15 Minutes extra time in the event  
of computer problems

**FACULTY OF  
ENGINEERING AND  
THE BUILT ENVIRONMENT**

**DEPARTMENT OF  
ELECTRICAL ENGINEERING**

**ES216BB  
ENGINEERING SOFTWARE DESIGN B**

**EVALUATION 2**

**SEPTEMBER 2026**

## EVALUATION INSTRUCTIONS

1. **Plagiarism:** Submit only original work. Similarity software will verify the authenticity of all submissions.
2. **Permitted Tools:** Use only CodeBlocks and Google Chrome to access the evaluation, view the paper, and upload your submission. Email, other online resources, memory sticks and unauthorised tools are prohibited.
3. **File Submission:** Rename the supplied source template to <student number>.cpp. Do not add a name, surname or other text to the file name.
4. **Uploading Instructions:** ONLY SUBMIT YOUR SOURCE FILE (.cpp) through the designated upload link. The latest submission is retained.
5. **Evaluation Scope:** This assessment covers Engineering Software Design B Units 1 to 4.
6. **Programming Language:** Construct the required program in C++11 and follow structured programming principles.
7. **Editing and Requirements:** Meet all specified requirements and use the attached appendices as required.
8. **Evaluation Requirements:**
  - a. Remember to save your work on the PC D: drive and save regularly throughout the evaluation.
  - b. Do not modify the given libraries, type declarations, prototypes, menus or comments in the supplied .cpp template.
  - c. Complete every required function in the comment blocks and use the exact names and parameters supplied in the question paper and template.
  - d. The supplied input file is **energy.txt**.

## C++ FILE CODE EXPLANATION

You are appointed to monitor the geyser energy consumption of a number of flats. You decide to implement a database, where the monthly energy consumption of all geysers is logged in a file. You then develop a C++ application to analyse the geyser data. The application sets up a menu-driven **Geyser Energy Management System**. It imports records from **energy.txt**, stores them in a linked list, displays the data, calculates the average geyser energy consumption, and deletes all pointers when done.

The program uses a structure named **TGeyserEnergy**.

The required type declarations and functions below must be implemented in the appropriate comment blocks in the supplied .cpp template file.

Use the exact function names and parameters from the supplied template.

## STRUCTURE DEFINITION EXPLANATION

The TGeyserEnergy structure and supporting types define the properties and operations required by this engineering system:

- **Structure and type requirements:**
  - **geyserID** (int): geyser identity.
  - **ratedPower** (float): geyser's rated power in kW.
  - **energyUsed** (float): geyser's energy consumed for the month in kWh.
  - **\*next** (TGeyserEnergy): pointer used in the linked list.
  - Some member functions
- **In the Main function:**
  - **startNode**: start node pointer to the TGeyserEnergy linked list containing geyser data.
- **Required functions:**
  - **InitialiseGeyser ()**: initialise one geyser record member function.
  - **AddGeyserToList ()**: add one geyser to the list.
  - **ReadDataFromFile ()**: create/reload dynamic data.
  - **DisplayGeyserData ()**: display scenario information.
  - **AverageEnergyConsumption ()**: calculate geysers' average energy consumption.
  - **DeleteSingleGeyser ()**: delete one geyser from the list.
  - **DeleteAllGeysers ()**: delete all geysers from the list.

- **File format:**
  - Each line in the file has the format:  
Geyser ID            Rated power            Energy used

---

**FUNCTION EXPLANATIONS**

---

**1. *InitialiseGeyser* Function**

***void TGeysersEnergy::InitialiseGeyser (int, float, float)***

**Purpose:** Initialises the structure variables with the geyser's data.

**Parameters:** geyser identifier, power, energy.

**Returns:** No value.

---

**2. *AddGeyserToList* Function**

***TGeysersEnergy\* AddGeyserToList(TGeysersEnergy \*\*startNode)***

**Purpose:** Adds one geyser to the existing list of geysers.

**Parameters:** double pointer to the start of the list.

**Returns:** a pointer to the new geyser added to the list.

---

**3. *ReadDataFromFile* Function**

***void ReadDataFromFile (string, TGeysersEnergy \*\*)***

**Purpose:** Deletes all existing geysers in the list (if there is already a list of geysers) before loading a new list of geysers from the file.

**Parameters:** file name, double pointer to the start of the list.

**Returns:** No value, updates the double pointer with the start of the list.

---

**4. *DisplayGeyserData* Function**

***void DisplayGeyserData (TGeysersEnergy \*startNode)***

**Purpose:** displays the list of geysers: id, rated power and energy consumption.

**Parameters:** pointer to the start of the list.

**Returns:** No value; outputs to the console.

---

**5. *AverageEnergyConsumption* Function**

***float AverageEnergyConsumption(TGeyserEnergy \*startNode)***

**Purpose:** calculate all the geysers' average energy consumption for the month.

**Parameters:** pointer to the start of the list.

**Returns:** average energy consumption.

---

**6. *DeleteSingleGeyser* Function**

***void DeleteSingleGeyser(TGeyserEnergy \*\*startNode)***

**Purpose:** delete a single geyser from the start of the list.

**Parameters:** double pointer to the start of the list.

**Returns:** No value. Updates the start of the list via the double pointer.

---

**7. *DeleteAllGeysers* Function**

***void DeleteAllGeysers(TGeyserEnergy \*\*startNode)***

**Purpose:** Delete the list of geysers (release pointers).

**Parameters:** double pointer to the start of the list.

**Returns:** No value; just cleans up the pointers. Updates the start of the list via the double pointer by setting it to the null pointer.

---

**PRINT SCREENS**

Main menu:

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: |
```

No data loaded:

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 2
```

No data loaded....

Press any key to continue...|

**Load geyser data:**

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 1

Geyser list created and Data from file loaded....

Press any key to continue...|
```

**Display geyser data:**

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 2

Geyser ID      Rated power (kW)      Energy consumption (kWh)
701             3.3                   432.1
702             3                     783.7
703             2.4                   250.8
704             3.1                   639.4
705             3                     157.5

Press any key to continue...|
```

**Display average geyser consumption:**

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 3

Average Geyser energy consumption: 452.7

Press any key to continue...|
```

**Delete single geyser (and displaying the list again with the geyser removed):**

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 4

First Geyser's information deleted....

Press any key to continue...|
```

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 2

Geyser ID          Rated power (kW)          Energy consumption (kWh)
702                 3                          783.7
703                 2.4                      250.8
704                 3.1                      639.4
705                 3                          157.5

Press any key to continue...|
```

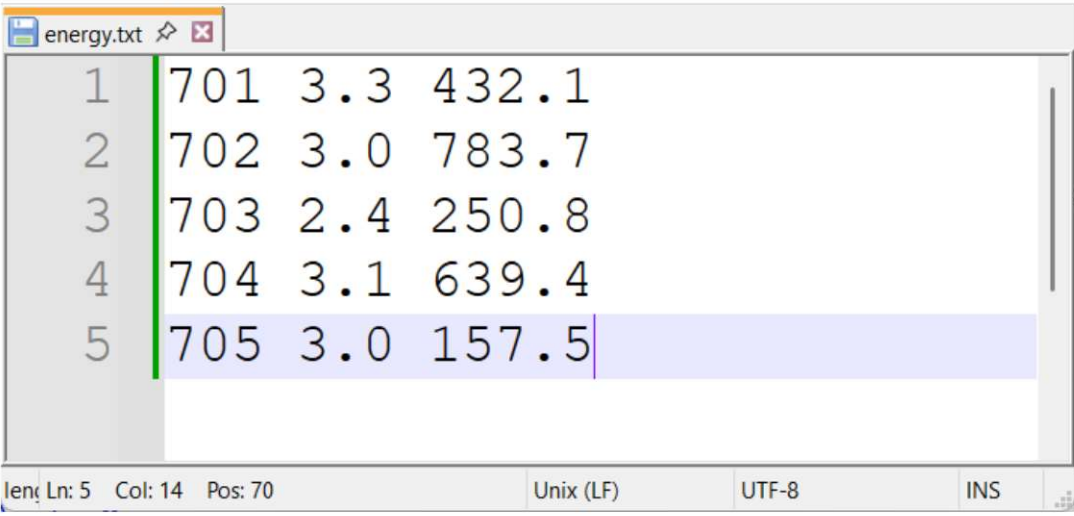
Exit application:

```
Geyser Energy Management System
ES216BB
Evaluation 2

1. Load Geyser information
2. Display Geyser information
3. Display average Geyser energy consumption
4. Delete a single Geyser's information
5. Exit
Choice: 5

Deleted all Geyser data, exiting program...

Press any key to continue...|
```

**Text File Input:**

The screenshot shows a text editor window titled 'energy.txt'. The editor contains five lines of text, each with four space-separated values. The fifth line, '705 3.0 157.5', is highlighted in light blue. A vertical green line is positioned at the start of the first column, and a vertical purple line is at the end of the fourth column. The status bar at the bottom indicates 'Ln: 5 Col: 14 Pos: 70', 'Unix (LF)', 'UTF-8', and 'INS'.

|   |     |     |       |
|---|-----|-----|-------|
| 1 | 701 | 3.3 | 432.1 |
| 2 | 702 | 3.0 | 783.7 |
| 3 | 703 | 2.4 | 250.8 |
| 4 | 704 | 3.1 | 639.4 |
| 5 | 705 | 3.0 | 157.5 |

Ln: 5 Col: 14 Pos: 70 Unix (LF) UTF-8 INS

**ANNEXURE A – MARK ALLOCATION**

*Note: Score range is 0 - 4: 0-none, 1-poor, 2-average, 3-good, 4-excellent*

| TEST RUBRIC                                     | SCORE [0-4] | WEIGHT [%] |
|-------------------------------------------------|-------------|------------|
| <b>C++ CODE EVALUATION</b>                      |             | <b>50</b>  |
| 0. Structures, types, static and dynamic access |             | 5          |
| 1. Function 1                                   |             | 5          |
| 2. Function 2                                   |             | 7          |
| 3. Function 3                                   |             | 8          |
| 4. Function 4                                   |             | 5          |
| 5. Function 5                                   |             | 5          |
| 6. Function 6                                   |             | 5          |
| 7. Function 7                                   |             | 5          |
|                                                 |             |            |
| 8. Compile and runtime stability                |             | 5          |
| <b>TOTAL</b>                                    |             | <b>50</b>  |

| Graduate Attribute                                                   | GA Number | GA Score [0-5] |
|----------------------------------------------------------------------|-----------|----------------|
| Application of scientific and engineering knowledge                  | GA2       | 0,1,4          |
| Engineering methods, skills, tools, including information technology | GA5       | 1,2,3,5,6      |
| Impact of Engineering Activity                                       | GA7       | 4,5,6          |
| Engineering Professionalism                                          | GA10      | 7,8            |

**ANNEXURE B – INFORMATION SHEET**

**Data types:** void, char, short, int, float, double

**Data Type modifiers:** const, auto, static, unsigned, signed

**Arithmetic operators:** \* / % + -

**Relational operators:** < <= > >= == !=

**Assignment operator:** = += -= \*= /= %= &= ^= |= <<= >>=

**Logic operators:** && || !

**Bitwise logic operators:** & | ^ ~ << >>

**Pointer operators:** Dereference: \* Address: &

**Control Structures:**

**IF Selection:** if (condition) { ... };

**IF ELSE Selection:** if (condition) { ... } else { ... };

**WHILE Loop:** while (condition) { ... };

**DO WHILE loop:** do { ... } while (condition);

**FOR Loop:** for (initial value of control variable; loop condition;  
increment of control variable) { ... }

**SWITCH Selection:** switch (control variable){ case 'value': ... ; break; default:  
... ; break; }

**Functions:** return\_data\_type function\_name ( parameters ) { ... };

**Arrays:**

One dimensional: data\_type variable\_name[size];

Two dimensional: data\_type variable\_name [x\_size][y\_size];

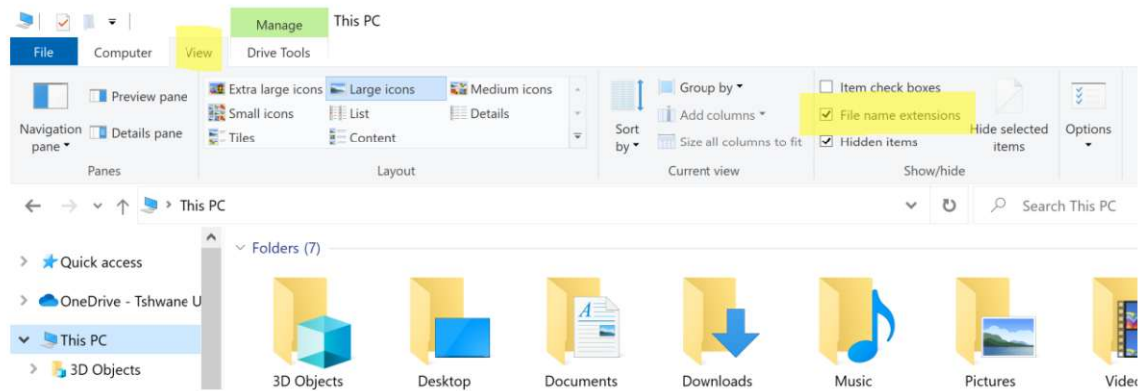
**ANNEXURE C – ASCII TABLE**

| Dec | Hx | Oct | Char                               | Dec | Hx | Oct | Html  | Chr          | Dec | Hx | Oct | Html  | Chr      | Dec | Hx | Oct | Html   | Chr        |
|-----|----|-----|------------------------------------|-----|----|-----|-------|--------------|-----|----|-----|-------|----------|-----|----|-----|--------|------------|
| 0   | 0  | 000 | <b>NUL</b> (null)                  | 32  | 20 | 040 | &#32; | <b>Space</b> | 64  | 40 | 100 | &#64; | <b>@</b> | 96  | 60 | 140 | &#96;  | <b>`</b>   |
| 1   | 1  | 001 | <b>SOH</b> (start of heading)      | 33  | 21 | 041 | &#33; | <b>!</b>     | 65  | 41 | 101 | &#65; | <b>A</b> | 97  | 61 | 141 | &#97;  | <b>a</b>   |
| 2   | 2  | 002 | <b>STX</b> (start of text)         | 34  | 22 | 042 | &#34; | <b>"</b>     | 66  | 42 | 102 | &#66; | <b>B</b> | 98  | 62 | 142 | &#98;  | <b>b</b>   |
| 3   | 3  | 003 | <b>ETX</b> (end of text)           | 35  | 23 | 043 | &#35; | <b>#</b>     | 67  | 43 | 103 | &#67; | <b>C</b> | 99  | 63 | 143 | &#99;  | <b>c</b>   |
| 4   | 4  | 004 | <b>EOT</b> (end of transmission)   | 36  | 24 | 044 | &#36; | <b>\$</b>    | 68  | 44 | 104 | &#68; | <b>D</b> | 100 | 64 | 144 | &#100; | <b>d</b>   |
| 5   | 5  | 005 | <b>ENQ</b> (enquiry)               | 37  | 25 | 045 | &#37; | <b>%</b>     | 69  | 45 | 105 | &#69; | <b>E</b> | 101 | 65 | 145 | &#101; | <b>e</b>   |
| 6   | 6  | 006 | <b>ACK</b> (acknowledge)           | 38  | 26 | 046 | &#38; | <b>&amp;</b> | 70  | 46 | 106 | &#70; | <b>F</b> | 102 | 66 | 146 | &#102; | <b>f</b>   |
| 7   | 7  | 007 | <b>BEL</b> (bell)                  | 39  | 27 | 047 | &#39; | <b>'</b>     | 71  | 47 | 107 | &#71; | <b>G</b> | 103 | 67 | 147 | &#103; | <b>g</b>   |
| 8   | 8  | 010 | <b>BS</b> (backspace)              | 40  | 28 | 050 | &#40; | <b>(</b>     | 72  | 48 | 110 | &#72; | <b>H</b> | 104 | 68 | 150 | &#104; | <b>h</b>   |
| 9   | 9  | 011 | <b>TAB</b> (horizontal tab)        | 41  | 29 | 051 | &#41; | <b>)</b>     | 73  | 49 | 111 | &#73; | <b>I</b> | 105 | 69 | 151 | &#105; | <b>i</b>   |
| 10  | A  | 012 | <b>LF</b> (NL line feed, new line) | 42  | 2A | 052 | &#42; | <b>*</b>     | 74  | 4A | 112 | &#74; | <b>J</b> | 106 | 6A | 152 | &#106; | <b>j</b>   |
| 11  | B  | 013 | <b>VT</b> (vertical tab)           | 43  | 2B | 053 | &#43; | <b>+</b>     | 75  | 4B | 113 | &#75; | <b>K</b> | 107 | 6B | 153 | &#107; | <b>k</b>   |
| 12  | C  | 014 | <b>FF</b> (NP form feed, new page) | 44  | 2C | 054 | &#44; | <b>,</b>     | 76  | 4C | 114 | &#76; | <b>L</b> | 108 | 6C | 154 | &#108; | <b>l</b>   |
| 13  | D  | 015 | <b>CR</b> (carriage return)        | 45  | 2D | 055 | &#45; | <b>-</b>     | 77  | 4D | 115 | &#77; | <b>M</b> | 109 | 6D | 155 | &#109; | <b>m</b>   |
| 14  | E  | 016 | <b>SO</b> (shift out)              | 46  | 2E | 056 | &#46; | <b>.</b>     | 78  | 4E | 116 | &#78; | <b>N</b> | 110 | 6E | 156 | &#110; | <b>n</b>   |
| 15  | F  | 017 | <b>SI</b> (shift in)               | 47  | 2F | 057 | &#47; | <b>/</b>     | 79  | 4F | 117 | &#79; | <b>O</b> | 111 | 6F | 157 | &#111; | <b>o</b>   |
| 16  | 10 | 020 | <b>DLE</b> (data link escape)      | 48  | 30 | 060 | &#48; | <b>0</b>     | 80  | 50 | 120 | &#80; | <b>P</b> | 112 | 70 | 160 | &#112; | <b>p</b>   |
| 17  | 11 | 021 | <b>DC1</b> (device control 1)      | 49  | 31 | 061 | &#49; | <b>1</b>     | 81  | 51 | 121 | &#81; | <b>Q</b> | 113 | 71 | 161 | &#113; | <b>q</b>   |
| 18  | 12 | 022 | <b>DC2</b> (device control 2)      | 50  | 32 | 062 | &#50; | <b>2</b>     | 82  | 52 | 122 | &#82; | <b>R</b> | 114 | 72 | 162 | &#114; | <b>r</b>   |
| 19  | 13 | 023 | <b>DC3</b> (device control 3)      | 51  | 33 | 063 | &#51; | <b>3</b>     | 83  | 53 | 123 | &#83; | <b>S</b> | 115 | 73 | 163 | &#115; | <b>s</b>   |
| 20  | 14 | 024 | <b>DC4</b> (device control 4)      | 52  | 34 | 064 | &#52; | <b>4</b>     | 84  | 54 | 124 | &#84; | <b>T</b> | 116 | 74 | 164 | &#116; | <b>t</b>   |
| 21  | 15 | 025 | <b>NAK</b> (negative acknowledge)  | 53  | 35 | 065 | &#53; | <b>5</b>     | 85  | 55 | 125 | &#85; | <b>U</b> | 117 | 75 | 165 | &#117; | <b>u</b>   |
| 22  | 16 | 026 | <b>SYN</b> (synchronous idle)      | 54  | 36 | 066 | &#54; | <b>6</b>     | 86  | 56 | 126 | &#86; | <b>V</b> | 118 | 76 | 166 | &#118; | <b>v</b>   |
| 23  | 17 | 027 | <b>ETB</b> (end of trans. block)   | 55  | 37 | 067 | &#55; | <b>7</b>     | 87  | 57 | 127 | &#87; | <b>W</b> | 119 | 77 | 167 | &#119; | <b>w</b>   |
| 24  | 18 | 030 | <b>CAN</b> (cancel)                | 56  | 38 | 070 | &#56; | <b>8</b>     | 88  | 58 | 130 | &#88; | <b>X</b> | 120 | 78 | 170 | &#120; | <b>x</b>   |
| 25  | 19 | 031 | <b>EM</b> (end of medium)          | 57  | 39 | 071 | &#57; | <b>9</b>     | 89  | 59 | 131 | &#89; | <b>Y</b> | 121 | 79 | 171 | &#121; | <b>y</b>   |
| 26  | 1A | 032 | <b>SUB</b> (substitute)            | 58  | 3A | 072 | &#58; | <b>:</b>     | 90  | 5A | 132 | &#90; | <b>Z</b> | 122 | 7A | 172 | &#122; | <b>z</b>   |
| 27  | 1B | 033 | <b>ESC</b> (escape)                | 59  | 3B | 073 | &#59; | <b>;</b>     | 91  | 5B | 133 | &#91; | <b>[</b> | 123 | 7B | 173 | &#123; | <b>{</b>   |
| 28  | 1C | 034 | <b>FS</b> (file separator)         | 60  | 3C | 074 | &#60; | <b>&lt;</b>  | 92  | 5C | 134 | &#92; | <b>\</b> | 124 | 7C | 174 | &#124; | <b> </b>   |
| 29  | 1D | 035 | <b>GS</b> (group separator)        | 61  | 3D | 075 | &#61; | <b>=</b>     | 93  | 5D | 135 | &#93; | <b>]</b> | 125 | 7D | 175 | &#125; | <b>}</b>   |
| 30  | 1E | 036 | <b>RS</b> (record separator)       | 62  | 3E | 076 | &#62; | <b>&gt;</b>  | 94  | 5E | 136 | &#94; | <b>^</b> | 126 | 7E | 176 | &#126; | <b>~</b>   |
| 31  | 1F | 037 | <b>US</b> (unit separator)         | 63  | 3F | 077 | &#63; | <b>?</b>     | 95  | 5F | 137 | &#95; | <b>_</b> | 127 | 7F | 177 | &#127; | <b>DEL</b> |

Source: [www.LookupTables.com](http://www.LookupTables.com)

**ANNEXURE D – HOW TO RUN THE SHOWCASE FILE**

1. Enable file extensions (see highlighted in yellow)



2. Change the name from “**Showcase.old**” to “**Showcase.exe**”
3. Run the “**ShowcaseEV.exe**” by double-clicking on the icon.
4. Windows may show the following. Click on “**More info**”



5. Click on “**Run anyway**”

